

Google Summer of Code Proposal: Empowering NixOS with First-Class Crystal Application Support

Project Title: Building a Robust Foundation for Crystal Applications in NixOS

Student: Vidhvath J

Skills: Nix, Bash, Crystal (or Ruby)

Mentor: @peterhoeg

Project Size: 500 hours

Rating: Easy

Project Summary

NixOS offers a compelling environment for building and deploying applications. However, existing support for Crystal applications requires improvement. This project aims to elevate Crystal application development within NixOS by:

- **Refactoring the Crystal Build Infrastructure:** Streamlining and improving the maintainability of the build infrastructure within `nixpkgs`.
- **Extending `crystal2nix` Tool Capabilities:** Expanding the `crystal2nix` tool to support dependency fetching beyond Git repositories.

Current State and Limitations

Currently, NixOS offers two key components for Crystal applications:

1. **`crystal2nix`:** This tool converts lock files generated by the `shards` tool (Crystal's package manager) into Nix expressions suitable for building Crystal applications within NixOS.
2. **`nixpkgs` Build Infrastructure:** Leveraging the outputs of `crystal2nix`, `nixpkgs` provides various functionalities for building Crystal applications.

However, the existing support faces limitations:

- **Brittle and Unreadable Code:** The current `nixpkgs` build infrastructure for Crystal applications is difficult to understand and maintain.
- **Limited Dependency Fetching:** The `crystal2nix` tool currently only supports fetching dependencies from Git repositories.

Expected Outcomes

This project aims to deliver significant improvements:

- **Enhanced Build Infrastructure:** A more user-friendly and maintainable build infrastructure within `nixpkgs` for Crystal applications, facilitating future development and contributions.
- **Expanded Dependency Management:** An extended `crystal2nix` tool capable of handling dependencies from various Version Control Systems (VCS) beyond Git, offering greater flexibility for developers.

Technical Approach

This project will be executed through a meticulously planned approach:

1. **In-Depth Code Analysis:** I will begin with a comprehensive analysis of the existing Crystal build infrastructure within `nixpkgs`.
2. **Refactoring for Maintainability:** Leveraging my understanding of Nix and Bash, I will refactor the codebase to enhance readability, modularity, and overall maintainability. This will facilitate future improvements and contributions to the codebase.
3. **Expanding `crystal2nix` Functionality:** I will collaborate with the mentor and the NixOS community to identify the most relevant VCS systems to target for dependency fetching. Subsequently, I will extend the `crystal2nix` tool to seamlessly handle fetching dependencies from these additional sources. This

might involve implementing support for specific VCS APIs or leveraging existing libraries within the NixOS ecosystem.

4. **Comprehensive Documentation:** Throughout the project, I will prioritize creating detailed documentation for both the refactored build infrastructure and the extended `crystal2nix` tool. These documents will serve as valuable resources for developers working with Crystal applications in NixOS.

Contribution to NixOS

This project offers several advantages to the NixOS community:

- **Empowering Crystal Developers:** A user-friendly and maintainable build infrastructure will encourage broader adoption of Crystal within NixOS, fostering a more vibrant development environment.
- **Enhanced Development Experience:** Developers will benefit from an improved Crystal development experience within NixOS, simplifying the process of building and deploying their applications.
- **Increased Flexibility:** Expanding dependency fetching beyond Git repositories in `crystal2nix` empowers developers to utilize a wider range of tools and resources when building Crystal applications using NixOS.

My GSoC Contribution

I am a highly motivated individual with a strong grasp of Nix and Bash scripting. Additionally, I am eager to learn Crystal; alternatively, my existing knowledge of Ruby (which is syntactically similar to Crystal) will be a valuable asset. I believe my skillset aligns perfectly with this project, and I approach it with the following commitments:

- **Close Collaboration with the Mentor:** I will actively seek guidance from @peterhoeg (if confirmed) to ensure the project aligns with the community's direction and best practices.

- **Community Engagement:** I will actively participate in discussions on the NixOS Discourse platform, learning from experienced developers and contributing to the overall project vision.
- **Testing and Continuous Improvement:** I will prioritize robust testing throughout the project to ensure all modifications function as intended. I will also be open to feedback and suggestions for further improvements from the NixOS community.

Prior Efforts and Project Viability

The existing `crystal2nix` tool and `nixpkgs` build infrastructure serve as a robust foundation for this project. My approach focuses on building upon these existing functionalities, enhancing their maintainability and capabilities for the NixOS community.

Conclusion

In conclusion, this project offers a clear path to enhance Crystal application development within NixOS. By refactoring the build infrastructure and extending `crystal2nix`, we can create a more user-friendly and flexible development environment. I am confident that with the skills and commitment I bring, this project will empower the NixOS Crystal community.